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A Distribution in Motion: The Case of Argentina

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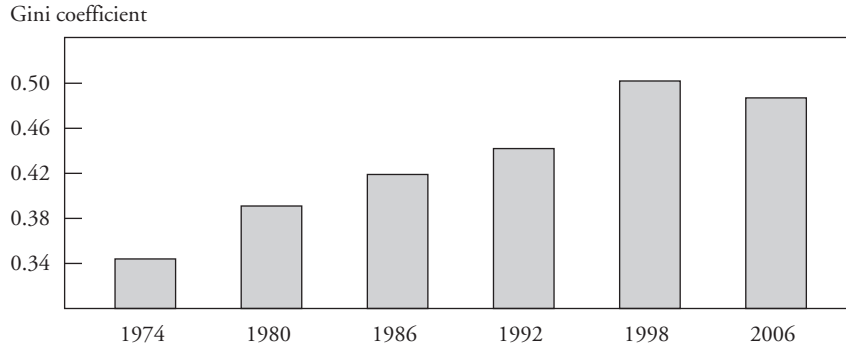
All of the available empirical evidence suggests that income inequality in Argentina substantially increased from the mid-1970s to the mid-2000s. Figure 5-1 presents a summary of this disappointing story: the Gini coefficient of the distribution of household per capita income for Greater Buenos Aires (GBA) soared from 0.344 in 1974 to 0.487 in 2006.[1] Inequality did not increase at a uniform rate over time, however. Figure 5-2 depicts short intervals of relative calm as

This is a shortened version of our paper for the UNDP project "Markets, the State, and the Dynamics of Inequality: How to Advance Inclusive Growth." This work benefited from detailed comments and suggestions by Nora Lustig and Luis Felipe López-Calva, the project directors, and from comments received at project seminars at UNDP headquarters in New York (November 2007) and at UNDP Mexico, Federal District (June 2008). We also wish to thank Alejandro Bonvecchi, Roxana Maurizio, and Martín Tetaz for their valuable input; Sebastián Etchemendy and Alberto Porto for discussions on the topics covered in this chapter's concluding remarks; and Mariellen Malloy Jewers for her excellent editorial suggestions. We are grateful to Sergio Olivieri and Gabriel Facchini (from CEDLAS) for outstanding research assistance.

1. The microdata behind the figures in this chapter come from Argentina's main official household survey, Encuesta Permanente de Hogares (EPH), which covers Argentina's major urban areas. The EPH started in the 1970s as a survey for Greater Buenos Aires, which accounts for one-third of Argentina's population. It was gradually extended, and since the early 1990s the EPH has covered all urban areas with more than 100,000 inhabitants, which represent two-thirds of Argentina's total population. The income concept used to measure inequality is monetary current per capita income. It does not include the value of imputed rent or auto-consumption or capital gains; it does include monetary transfers, both private and public. The EPH does not make explicit whether reported income is before or after taxes and social security contributions, but the authors presume that for salaried workers, income reported in the EPH is probably net of

Figure 5-1. *Distribution of Household Per Capita Income, Greater Buenos Aires, 1974–2006*



Source: Authors' calculations based on microdata from the Encuesta Permanente de Hogares (EPH) (INDEC, various years).

well as intervals of surges in inequality.[2] Although an inequality series from the mid-1970s can be estimated only for Greater Buenos Aires, which contains roughly one-third of Argentina's total population, the trends described can be extrapolated to the whole urban population. During the period from the mid-1970s to the mid-2000s, the increase in inequality in Argentina was significantly larger than increases in inequality in most other countries worldwide (Gasparini, Gutiérrez, and Tornarolli 2007). The difference between the extent of inequality in Argentina and more unequal economies, like those of Brazil, Chile or Mexico, shrank substantially in the last couple of decades. Argentina's once proud "European" income distribution, with a large middle class, had moved closer toward a "Latin American" income distribution, with high levels of inequality.[3]

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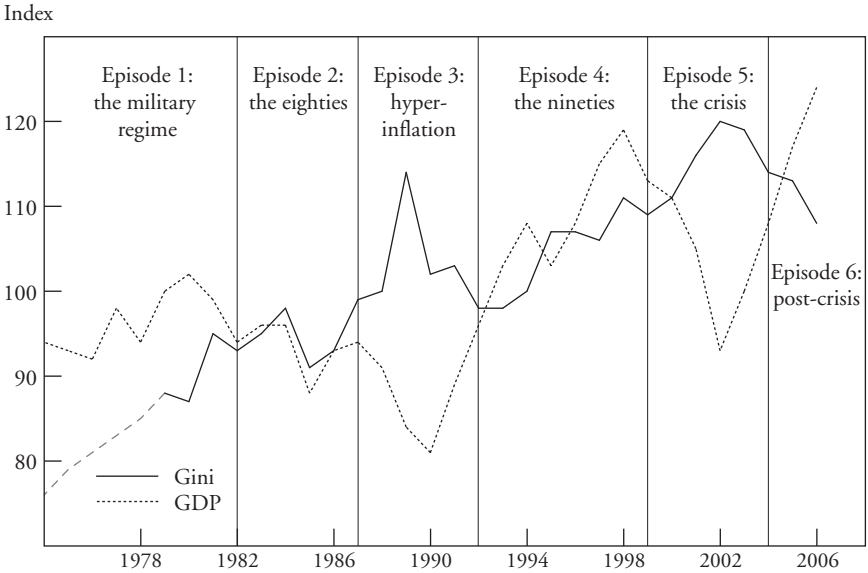
both. Income data were not adjusted for nonsampling errors, such as underreporting. The appendix of our companion paper (Gasparini and Cruces 2008) provides more detailed information about the methodology and coverage of the EPH.

2. This upward trend in inequality is robust to the geographic coverage of the data source, to the choice of indicator, and to methodological aspects regarding the underlying income variable. The trend also is apparent using alternative data sources. The upward trend in inequality in household per capita income is also robust to the choice of indicator. The confidence intervals for the Gini coefficient show that observed changes in inequality are, for the most part, statistically significant. Moreover, Gasparini (2005b) shows that this trend is robust to a host of methodological issues in household surveys, including nonresponse, misreporting of income, inclusion of nonmonetary income, inclusion of implicit rent from homeownership, accounting for family structure through equivalization, and adjustment for regional prices, among others. The trends also are similar when including or excluding the impact of social expenditure and taxes on household income.

3. We know that surveys underestimate top incomes. Argentina is at present the only Latin American country for which we can have an idea of the extent to which top incomes are underestimated. In chapter 4, Alvaredo and Piketty, using data from tax returns, estimate that the Gini for 2004 could have ranged between .52 and .57, depending on whether it is assumed that the top .1 percent or the top 1 percent,

Figure 5-2. *Variation in Gini Coefficient of Distribution of Household Per Capita Income and Per Capita GDP, 1974–2006*[a]



Source: Authors' calculations based on microdata from the Encuesta Permanente de Hogares (EPH) (INDEC, various years).

a. Mean index = 100.

This chapter focuses on early 1990s and the mid-2000s, a period characterized by a sharp increase in inequality until the early 2000s and a decline thereafter. The period covers two contrasting economic policy regimes. In the 1990s, Argentina went through far-reaching, market-oriented reforms in a context of weak labor market institutions and limited social protections. In the 2000s, state intervention in the economy became more pervasive, labor market institutions were stronger, and social protection schemes redistributed income to unskilled and semi-skilled workers.

The chapter begins with a presentation of the main trends in Argentina's income distribution from the mid-1970s to the mid-2000s in terms of six distinct historical episodes, then explores the changes in inequality during noncrisis years by means of microeconometric and sectoral decompositions. Next it examines the relative contribution of three major factors in Argentina's evolution of inequality—trade liberalization, technological change, and macroeconomic adjust-

respectively, were not included in the survey data. The survey-based Gini was estimated at .49. The trend for the series with top incomes differs slightly from that for the series based on household surveys between 2001 and 2004. While both show a declining trend, the magnitude is small or negligible in the series with top incomes (see table 4-1 in Alvaredo and Piketty).

ment—and covers other contributing factors for which empirical analysis is scarce, such as demographic factors, privatization, and unionization. The chapter then investigates the distributional impact of social spending and its financing through the tax system; examines poverty and inequality during macroeconomic crises; and analyzes the decline in inequality experienced during Argentina's economic recovery following the 2001–02 crisis.

An Episodic Story of Inequality

As stressed by Atkinson (1997), inequality changes can be analyzed from a long-term perspective or by dividing time into episodes. The episodic story of Argentina's changes in inequality seems to be more compelling than the long-term story. One reason for choosing an episodic story is that intense economic shocks and policy changes have affected Argentina's income distribution in different ways. Likewise, the logic behind Argentina's changes in inequality is different at different intervals of time. Therefore, taking a long-term perspective would miss much of the action and probably would be unhelpful in thinking about the future.

The first episode starts in 1974, since data from the Encuesta Permanente de Hogares (EPH) are available from that year onward, and the following 32 years are divided into six episodes (see table 5-1).[4] The first episode covers the years from 1974 to 1982, encompassing the last two years of a democratic government (1974–76) and an entire dictatorial military regime. The first episode is characterized by several factors: weak labor institutions, with almost no role for unions; a sweeping trade liberalization reform; and a sharp overall increase in inequality. The second episode comprises most of the 1980s (1983–87), and it is characterized by other factors: the return to democratic rule; a substantially more closed economy; increased union activity and stronger labor institutions (minimum wage enforcement, collective bargaining); macroeconomic instability; and a rather stable income distribution. The third episode (1988–91) corresponds to the macroeconomic crisis of the late 1980s, which included two hyperinflations. This episode is characterized by a sharp increase and a subsequent sudden fall in inequality after the economy stabilized in 1991. The fourth episode, which includes most of the 1990s (1992–99), is characterized by several factors: relative macroeconomic stability;[5] a currency board with an exchange rate fixed to the

4. In March 1976, by means of a coup d'état, a military regime came into power. Although, arguably, the first episode should start at that point, information from the EPH is available first for 1974, and then from 1980 onward. Episode 1 thus starts in 1974, although most of the observed distributional changes are attributed to developments under the military regime.

5. This stability refers mainly to the curbing of inflation, which was linked to the fixed exchange rate regime (currency board) set in place in 1991. The opening of the economy to capital flows implied a high degree of exposure to international fluctuations and to flow reversals, as witnessed by the impact of the successive crises in Mexico, Southeast Asia, Russia, and Brazil. See the discussion of macroeconomic crises below for more details.

Table 5-1. *Episodes*

Factor	*Episode 1 Military regime 1974–82*	*Episode 2 The 1980s 1983–87*	*Episode 3 Hyperinflation 1988–91*	*Episode 4 The 1990s 1992–99*	*Episode 5 The crisis 2000–04*	*Episode 6 Post-crisis 2005–06*
Macro situation	Low growth and crisis	Stagnation	Crisis and recovery	Growth	Crisis and recovery	Growth
Import competition	High (not always)	Low		High		Low
Technological change	Low	Low		High		Moderate
Labor institutions	Weak	Strong		Weak		Strong
Social protection (cash)	Low	Low	Low	Low	High since 2002	Moderate
Inequality	Increase	Stable	Increase and fall	Increase	Increase and fall	Fall
Poverty	Stable	Increase	Increase and fall	Increase	Increase and fall	Fall

Source: Authors' calculations based on microdata from the Encuesta Permanente de Hoyares (EPH) (INDEC, various years).

U.S. dollar; deep structural reforms, which implied a much more open and flexible economy; and weaker labor institutions. Finally, in the fourth episode, the income distribution during the 1990s became substantially more unequal than it ever had been in Argentina's history. The fifth episode, covering the period from 2000 to 2004, includes the deep macroeconomic crisis of 2001–02, which was followed by an economic meltdown and the devaluation of the currency. The fifth episode, similar to the fourth, is characterized by a sharp increase in inequality and a subsequent substantial decline in inequality after economic stabilization. The sixth episode, under way at the time of writing, started around 2005 with rapid growth in the aftermath of the macroeconomic crisis of 2001–02. This episode's main characteristics include the adjustment of economic agents to the new relative prices introduced by Argentina's currency devaluation in 2002, stronger labor institutions, and a more extensive safety net than in prior episodes. Inequality fell to precrisis levels during this episode. Figure 5-2 reproduces the pattern of the Gini coefficients and GDP per capita marking the six episodes.[6]

Like any other modeling exercise, the episodic story tries to highlight the main aspects of a very complex stream of phenomena.[7] Table 5-1 characterizes the episodes in terms of five elements: macroeconomic situation; import competition; technological change and physical capital accumulation; unions and labor institutions; and social protection. Naturally, changes in Argentina's income distribution are the result of a vast array of factors, so any simple classification excludes potentially relevant explanations. The five elements in table 5-1 were chosen because they have two important qualities in common: according to theory, they are closely linked to changes in income distribution, and they have been extensively reviewed in the distributional literature on Argentina.

The six proposed episodes can be classified into three types: type-1 episodes, distinguished by serious macroeconomic crisis (episodes three and five); type-2 episodes, distinguished by liberalization with weak labor institutions (episodes one and four); and type-3 episodes, distinguished by low import penetration and stronger labor institutions (episodes two and six). Inequality seems to have fluctuated widely in type-1 episodes, increased in a rather permanent way in type-2 episodes, and decreased or remained stable in type-3 episodes.

A word of caution is necessary before proceeding with the analysis. It is not easy to derive policy conclusions even from this stylized story, since it is very difficult

6. In chapter 3 of this volume, James Robinson discusses how the dynamics of income distribution in Argentina over the entire twentieth century are closely related to movements backward and forward between authoritarian and democratic regimes. While the type of regime undoubtedly influenced the dynamics of inequality in particular through the weakening or strengthening of labor unions, the relationship is not a one-to-one relationship by far. Robinson also notes that in Argentina there have been episodes in which inequality increased under democratic rule.

7. This episodic review is necessarily simplified, and some aspects, in particular the limits between episodes, are somewhat arbitrary. The reader is referred to Gerchunoff and Llach (2004) and the references therein for a complete account of the period under discussion.

to assess the long-run general equilibrium effects of policies. For example, a given combination of policies or circumstances may be associated with low inequality and poverty in the short run but might contribute substantially to a future crisis in which all the distributional improvements are undone. In that sense, the above classification and the analysis that follows are mostly descriptive.

Exploring the Proximate Determinants of Inequality Changes

Examining only noncrisis years allows one to assess the relevance of proximate factors—that is, not to take into account behavioral changes—in Argentina's inequality trend. We begin by showing the results of a microdecomposition that highlights the main factors contributing to the increase in inequality from 1992 to 1998. Next, we present evidence of a widening gap between the wages of skilled workers and those of unskilled workers, which is a key factor in the surge in inequality. We end with an assessment of the impact of labor supply and sectoral changes on the wage gap.

Microdecomposition Exercises

The first exercise consists of a microeconometric decomposition of the changes in the Gini coefficient for the distribution of hourly wages, earnings, and household income. The methodology closely follows that of Gasparini, Marchionni, and Sosa Escudero (2004). The basic idea of the microsimulations is to find the counterfactual distribution of individual earnings that would be generated in a given period t_1 if some of the determinants of earnings took the observed values in t_2 and the rest remained at their values in t_1. The difference between the real distribution in time t_1 and the counterfactual distribution in time t_2 is equal to the distributional impact of the factors modified in the simulation.[8] By modeling the determinant of wages and hours worked, the microsimulations incorporate behavioral factors into the analysis, at the cost of sometimes having substantial unexplained residuals.

Table 5-2 presents the results for Greater Buenos Aires, which is the only region for which data that go back to 1980 are available.[9] The table shows the impact of six counterfactual exercises on the distribution of hourly wages, earnings, and household income. The simulations include changes in the returns to education in terms of hourly wages; the returns to education in terms of hourly wages; the gender wage gap; the distribution of unobservable factors (for example, ability, connections, quality of education); the determinants of employment; and the distribution of education. For each period, table 5-2 reports the actual

8. Since the outcomes are path-dependent, the tables corresponding to the microsimulations report the mean of the results obtained by changing the base year in each simulation.

9. It is difficult to extend the microsimulation to 1974, since the data set for that year includes fewer variables.

change in the Gini coefficient, along with six counterfactual changes corresponding to various simulated effects. For example, column 2 shows the change in the Gini resulting from changing only the returns to education in terms of hourly wages (that is, the coefficients of the educational dummies in the hourly wage equation), keeping everything else constant. The change in the coefficients of the educational dummies in the hourly wage equation has a direct effect on the counterfactual distribution of hourly wages. Naturally, changing the educational dummies' coefficients also affects earnings and hence equivalized income.

Inequality in hourly wages and earnings diminished in the 1980s (ignoring the macroeconomic crisis of the late 1980s), driven by a fall in the returns to education in terms of hourly wages. In contrast, during the 1990s the returns to education in terms of hourly wages contributed to higher inequality in hourly wages and earnings. From 1992–98 the overall effect of the returns to education on the distribution of hourly wages and earnings accounted for 4.6 points of the 8.4 point increase in the Gini for the equivalized household income distribution from 1992–98. In addition, unobservable factors were responsible for 1.5 points of the increase in the Gini. Those results suggest that unskilled workers—both in terms of formal education and in terms of unobservable factors—lost relatively more than skilled workers in terms of hourly wages and hours of work during the 1990s. Any story of inequality changes in Argentina should pay particular attention to that phenomenon.

The decomposition can also shed some light on the debate on the contribution of unemployment to the surge in inequality. The increase in inequality in the 1990s occurred simultaneously with unprecedented growth in unemployment. The unemployment rate rose substantially—from around 3 percent in the 1970s to 15 percent in the late 1990s—in years of relative macroeconomic stability. According to figure 5-3, the labor market was rather quiet until the 1990s, when two combined phenomena generated a shock to the unemployment rate: a significant increase in the labor participation rate (from 40 percent in 1990 to 46 percent in 1999) and a fall in the employment rate driven by the macroeconomic shocks and the adjustments after various structural changes. Nonetheless, by the end of the 1990s the employment rate recovered, meaning that most of the increase in unemployment rate can be accounted for by the large increase in labor market participation. Women and youths joined the labor market in large numbers but faced an economy with a rigid employment rate.

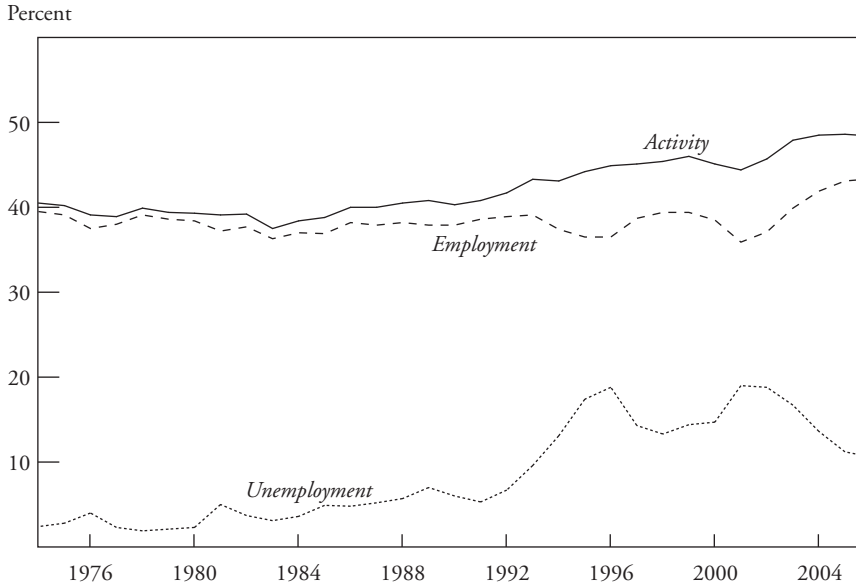
When unemployment is mostly a consequence of increased labor market participation instead of falling employment, its effect on inequality or poverty is less obvious. For instance, if a youth enters the labor force but is unable to find a job, the unemployment rate goes up but the income distribution remains unchanged. The direct distributional impact of employment issues can be assessed by means of counterfactual distributions derived from changing the coefficients of the variables in a labor participation equation. As reported in column 6 of table 5-2, this

Table 5-2. *Changes in the Gini Coefficient between periods, Actual and Simulated*

Period	Actual change (1)	Returns to education on: Wages (2)	Returns to education on: Hours worked (3)	Gender wage gap (4)	Unobservables (5)	Employment (6)	Education (7)	Rest (8)
1980–86								
Hourly wages	–1.5	–2.2		0.7	–2.8		1.1	1.7
Earnings	0.7	–1.4	0.0	1.0	–1.6	0.2	1.3	1.3
Equivalized income	3.1	–2.1	0.5	0.0	–1.6	–0.1	0.6	5.7
1986–92								
Hourly wages	–2.5	–1.6		–0.6	0.5		0.0	–0.8
Earnings	–1.7	–1.1	–0.4	–1.1	0.5	0.0	0.0	0.4
Equivalized income	0.7	0.9	–1.0	0.1	0.4	0.0	0.1	0.0
1992–98								
Hourly wages	5.5	3.0		–0.4	2.6		0.8	–0.5
Earnings	7.1	2.3	2.5	–0.5	1.8	–0.1	0.8	0.1
Equivalized income	8.4	2.7	1.9	0.0	1.5	0.2	0.7	1.4
1998–2006								
Hourly wages	–1.1	0.2		0.1	0.6		0.6	–2.6
Earnings	–1.5	0.1	–1.0	0.3	0.3	0.3	0.6	–2.0
Equivalized income	–1.7	–0.9	0.1	–0.2	0.3	–0.2	0.5	–1.4

Source: Authors' calculations based on microdata from the Encuesta Permanente de Hoyares (EPH) (INDEC, various years).

Figure 5-3. *Labor Market Indicators*



Source: INDEC (various years).

employment effect has been small, even during the years of increasing unemployment (1992–98). As discussed, this exercise provides just an estimate of the *direct* effect of changes in the extensive margin of the labor market; we elaborate on the possible indirect effects of a higher unemployment rate on wages later in the chapter.

Supply Factors

In a typical simple equilibrium model, the wage gap between skilled and unskilled workers is driven by changes in their relative supply and demand. The evidence reveals a strong increase in the relative supply of semi-skilled (high school graduates) and skilled (college graduates) workers, compared with unskilled workers (those with less than a high school degree). While 78.6 percent of adults aged 20 to 65 years were unskilled in GBA in 1974, that share fell significantly, to 47.1 percent, in 2006. In contrast, from 1974 to 2006 semi-skilled workers' share of adults aged 20 to 65 years in GBA climbed from 17.6 percent to 37 percent and skilled workers' share grew from 3.8 percent to 15.9 percent. Those patterns are valid for all urban areas in Argentina. Furthermore, the increases in semi-skilled and skilled workers are even more pronounced when the share in employment or in aggregate labor (total weighted hours of worked) is considered.

Given the strong increase in the relative supply of college graduates, the wage premium would have been bound to fall if factor demand had not changed. In

fact, that appears to have happened in the 1980s, though not in the 1990s, when the college wage premium jumped instead. That suggests an increase in the demand for skilled workers (especially for college graduates), which more than offset the downward pressure from the increased supply of skilled workers.

Sectoral Changes

Argentina's economy experienced large changes in its productive and employment structure over the period under study (1974–2006). The most noticeable change in the labor structure since the 1970s has been the fall in employment in the manufacturing industry and the increase in skilled services (public sector as well as professional and business services). From 1974 to 2006, employment in manufacturing industries dropped from 39 percent to just 17 percent. On the other hand, the share of employment in the more skill-intensive sectors of professional and business services and the government increased from 21 percent in 1974 to 41 percent in 2006. Those trends do not vary substantially when the population of workers is divided by skill level.

With skilled-biased technological change, the increase in the stock of more educated workers can be easily absorbed in each sector. In contrast, with constant production technologies, increases in skilled labor can be absorbed by either the relative growth of skilled-labor-intensive sectors, with no changes in the intensity of factor use, or an increase in the use of skilled workers in unskilled tasks. The evidence presented in our companion paper (Gasparini and Cruces 2008) is consistent with Argentina's having experienced a strong increase in the intensity of use of skilled labor in most sectors of the economy (that is, skill-biased technological change). The skill upgrading in production processes was especially strong in basic and high-tech manufacturing sectors and also in commerce and public administration.

To further explore the potential distributional impact of changes in employment, table 5-3 presents an update of the decomposition in Gasparini (2005a) of changes in the share of each type of labor in total aggregate labor. Formally, the table reports

$$\Delta\left[\frac{N_i}{N}\right] = \underbrace{\sum_s \frac{N_{ist}}{N_{st}} \Delta\left[\frac{N_s}{N}\right]}_{\text{Between effect}} + \underbrace{\sum_s \frac{N_{st'}}{N_{t'}} \Delta\left[\frac{N_{is}}{N_s}\right]}_{\text{Within effect}},$$

where N is labor input (number of workers multiplied by efficiency hours of work), i indexes the type of labor (unskilled, semi-skilled, and skilled), s indexes the economic sectors, and t indexes time. The first term of the decomposition (the "between" effect) captures the impact of transformations in the structure of employment sectors (for example, changes in the production structure driven by

Table 5-3. *Sectoral Decomposition of Changes in Labor*

Education	*1974–80*	*1980–86*	*1986–92*	*1992–98*	*1998–2006*
Between effect					
Less than high school	–1.6	–0.5	–3.8	–1.9	–0.3
High school graduate	0.5	0.3	0.8	0.6	–0.2
College graduate	1.1	0.2	3.0	1.3	0.6
Within effect					
Less than high school	–2.2	–7.3	–5.6	–5.4	–6.9
High school graduate	1.8	4.4	2.0	–0.3	4.9
College graduate	0.4	2.9	3.6	5.7	2.0
Overall effect					
Less than high school	–3.8	–7.8	–9.4	–7.3	–7.3
High school graduate	2.4	4.7	2.8	0.3	4.7
College graduate	1.4	3.1	6.6	7.0	2.6

Source: Authors' calculations based on microdata from the Encuesta Permanente de Hoyares (EPH) (INDEC, various years).

trade liberalization) on the relative employment of factor *i*. The second term (the "within" effect) captures changes due to variations in the intensity of use of different types of labor within each sector (for example, changes due to biased technological change).

Over time, changes across sectors benefited skilled workers primarily (and slightly benefited semi-skilled workers). The between effect for college graduates is always positive (table 5-3). In contrast, the between effect for unskilled workers is negative; they suffered from employment reallocations away from unskilled-labor-intensive sectors. The numbers in the second panel of table 5-3 suggest that the changes within sectors implied increased use of skilled labor. The highest within effect for college graduates is for the 1990s, a fact consistent with the story of capital incorporation and skilled-biased technological change after Argentina's reforms during that time. It is interesting to note that the values in the second panel are generally higher than those in the first panel, suggesting that within effects have been stronger than between effects.

Rising Inequality in the 1990s

Here we deepen the analysis of inequality changes in Argentina by studying the three main hypotheses that have been laid out in the inequality literature: trade liberalization, technological change and capital incorporation, and macroeconomic adjustment. In each case we start by sketching the main arguments linking these phenomena with inequality and then summarize the contributions that provide empirical evidence of those links.

Trade Liberalization

Although Argentina witnessed a series of reforms in almost all aspects of economic life from the mid-1970s to mid-2000s, most of the literature highlights trade liberalization as one of the key factors behind the increase in Argentina's income inequality. While that is due in part to the constraints on data availability for testing competing hypotheses, this line of research is motivated also by the salience of trade in the debate on increasing returns to skills in developed economies (Katz and Autor 1999).

The conventional wisdom in economic theory and some policy circles seemed to be that unskilled labor, a relatively abundant factor in developing economies, would benefit from trade reform and that thus inequality would fall in developing countries with increased trade liberalization (Perry and Olarreaga 2006). However, as documented in the thorough reviews in Goldberg and Pavcnik (2004, 2007), reality seldom corresponded to that "naïve view" of the equalizing impact of trade reform in developing countries. A host of reasons for the discrepancy between the naïve view and reality are described below.

In light of Argentina's status as a middle-income country and its abundance of natural resources, which are complementary to capital and skilled labor,[10] the impact of trade liberalization on inequality in Argentina was not clear-cut ex ante. However, the evidence for Argentina suggests overwhelmingly that trade liberalization led to an increase in inequality. Galiani and Sanguinetti (2003) was among the first studies to find evidence of an unequalizing effect of trade reform during the 1990s. The authors found that in those sectors in which import penetration was deeper, the wage gap between skilled and unskilled workers increased; however, this effect can explain only a minor fraction of the increase in inequality over the period. Galiani and Porto (2008) found that from 1974 to 2001, trade liberalization episodes in Argentina reduced wages on average and that reductions in industry tariffs increased the wage premium for skilled workers. Taken together, this evidence suggests that trade liberalizations increased skill premiums and thus contributed to higher overall income inequality in Argentina.

The general conclusion from the studies on the distributive impact of trade liberalization in Argentina is that more openness produced a wider gap between the wages of skilled and those of unskilled workers and thus higher levels of overall inequality. However, its impacts on the wage premium were small and can explain only a relatively small fraction of the total increase in the wage premium.

10. The relevant factors of production and trade partners that determine Argentina's relative abundance in the international trade arena is not a settled issue. Galiani and Sanguinetti (2003) argues that Argentina is abundant in unskilled labor compared with major trade partners like the United States and the European Uniion. Galiani and Porto (2008) highlights that Argentina is well-endowed in skills relative to the other countries in Latin America with which it trades. Cristini (1999) and Keifman (2006) consider Argentina a country abundant in land and skilled labor, while Berlinski (1994) shows specialization in natural resources and skilled labor when Argentina is compared with Brazil, its major partner within the region.

Technological Change and Capital Incorporation

A complementary explanation for the large fall in the relative demand for unskilled workers relies on the importance of skilled-biased technological change (SBTC) and capital incorporation in Argentina. Technological and organizational changes that increase the relative productivity of skilled workers translate into wider wage gaps. In addition, with labor market rigidities, those changes also translate into lower employment for unskilled workers. An increase in the use of physical capital in the production process increases inequality through two channels. First, if capital goods incorporate embedded technological change, an increase in investment in new machinery and equipment can accelerate the adoption of new technologies. Second, even without technological innovation, physical capital usually is more complementary to skilled labor, being then a source of an increasing productivity gap across workers with different education levels.

A SHOCK IN PHYSICAL CAPITAL, TECHNOLOGY, AND ORGANIZATION. For decades Argentina was a relatively closed economy with low investment rates; in the 1990s, however, a new scenario emerged. Argentina was experiencing macroeconomic stability and had instituted a set of market-oriented policies, including a massive process of privatization and deregulation, along with measures leading toward capital account liberalization. In addition, appreciation of the real exchange rate and large tariff reductions substantially reduced the relative price of physical capital and favorable international financial conditions contributed to the massive inflow of foreign capital to Argentina.

Private investment as a proportion of GDP increased 44 percent between the 1980s and the 1990s. In particular, foreign direct investment as a share of GDP increased from an average of 0.4 percent of GDP during 1970–90 to 1.6 percent in from 1991–97. The physical capital stock (excluding the public sector) grew by 20 percent between 1992 and 1999, and the average age of capital stock decreased from 8.8 to 5.2 years. The rapid increase in physical capital, particularly of imported machinery and equipment, was a vehicle of technological modernization in Argentina after decades of backwardness.

The deregulation of many domestic markets and the removal of barriers to international trade forced private firms to seek the productivity gains necessary to stay in business. In addition, the increased openness of the Argentine economy occurred at a time of increasing globalization and diffusion of new communication and information technologies, inducing firms to adopt state-of-the-art production technologies. Production processes in many sectors underwent radical changes, incorporating information technology, computers, robots, and modern assembly lines within just a few years.

Changes took place not only at the production stage but also at the organizational level. The scenario described above contributed to an extraordinary transformation in the property structure of firms—from public to private, from

domestic to foreign, and from small owners to large owners. The new structure favored changes toward larger production scales (for example, from small grocery stores to large supermarkets) and toward a more efficient use of production factors (for example, eliminating excess workers in the public sector).

Both technological and organizational changes implied lower relative demand for unskilled and semi-skilled workers than for skilled workers. Had those changes been adopted gradually or in the context of strong social protections, the impacts on unskilled and semi-skilled workers would have been milder. But that was not the case; the modernization of Argentina's economy took place in just a few years in the context of weak labor institutions and in the midst of labor deregulation.

THE EVIDENCE. One of the main arguments in favor of the technological change explanation is the difficulty of finding alternative plausible hypotheses to account for the co-movements of relative wages and skill intensity. As already mentioned, there was a substantial increase in the wage premium in the 1990s alongside an increase in the relative use of skilled labor across all economic sectors. Those observations are hard to reconcile with the predictions of most models that ignore skill-biased technological change.[11]

The sectoral decomposition of changes in the share of employment by educational groups described earlier in the chapter suggests that the fall in the relative employment of unskilled labor is accounted for mainly by a drop in the intensity of use of unskilled labor within all economic sectors. This "within" effect is especially relevant from 1992 to 1998, and it is consistent with the story of technological/organizational shock in the 1990s. García Swartz (1998) shares that conclusion, although the author's analysis covers just the beginning of the reforms.

The changes in the returns to education and unobservable factors documented previously also are compatible with the SBTC/capital accumulation hypothesis. First, in Argentina both the returns to education and unobserved skills increased substantially in the 1990s (and not in the 1980s), a fact that is consistent with a technological shock driving changes in both returns. Second, the change in the returns to education in Argentina implied substantial and increasing gains for college graduates compared to the rest of the population, without any clear change in the gap between unskilled and semi-skilled workers. That is compatible with a pattern in which a new technology is strongly complementary to nonroutine cognitive tasks (typical of highly skilled workers) and mostly a substitute for the routine tasks found in many traditional middle-wage jobs (like those in the manufacturing sector typical of semi-skilled or unskilled workers).

Acosta and Gasparini (2007) presents evidence for the relationship between capital accumulation and the wage structure by taking advantage of the variability of wage premiums and capital investment across industries in Argentina's man-

11. Explanations based on skilled workers performing unskilled tasks, which are consistent with these co-movements, are treated later in the chapter.

ufacturing sectors. The authors find that sectors that accumulated more physical capital in the 1990s were those in which the wage premium grew the most. Using a technology survey with longitudinal data, Bustos (2006) assesses the impact of trade and foreign investment on technology and skill upgrading at the firm level. The study shows that firms that upgraded technology faster also upgraded their need for skilled workers faster.

As discussed, the direct effect of trade liberalization on wage inequality seems to have been small. However, as argued above, it is likely that the profound trade and capital account liberalization process was key in fostering the rapid adoption of new technologies through various channels (see Acemoglu 2003, Atolia 2007, and Yeaple 2005).

The greater relevance of the capital/technology and trade/technology channels over the "pure" trade channel is found also in studies for other countries. Behrman, Birdsall, and Székeley (2003) combines policy indices with household survey microdata on wage differentials by schooling level for eighteen Latin American countries. The authors fail to find a significant effect of trade reform on wage differentials, but they do find an impact of the share of technology exports, which they use as a proxy for technology adoption, on wage differentials. Sánchez-Páramo and Schady (2003) reaches a similar conclusion using repeated cross-sections of household surveys for a series of Latin American countries. The authors stress an important point: although the direct effect of trade on wage inequality may be small, trade is an important mechanism for technology transmission.

Macroeconomic Adjustment

We have highlighted the relevance of the increased skill premium to wage inequality in Argentina and discussed the relevance of trade liberalization, capital incorporation, and technological change as plausible causes behind the increase in wage inequality and the ensuing increase in income inequality. Another strand of the literature sees the impact of macroeconomic adjustment and the resulting reduction in the aggregate demand for labor as a central argument for the increase in inequality in the 1990s.

That argument states that under falling aggregate demand for labor and an increasing supply of skilled workers, more educated workers take the occupations of unskilled workers, who are displaced and become either unemployed or underemployed. That adjustment reduces the mean wage of the skilled workers through a composition effect. However, at the same time, the labor market becomes even tougher for unskilled workers, who lose their jobs or work fewer hours, hence raising inequality. The higher intensity of use of skilled workers in production is then attributed to skilled workers displacing unskilled workers and not to the higher productivity of skilled labor. That argument is often linked to the concept of "credentialism": the same occupations are now performed by more qualified workers, although they have the same job description and basically the same productivity.

Maurizio (2001) finds some evidence of that phenomenon in Argentina, which the author argues was behind the increase in the educational level of the labor force in the 1990s.[12] The finding of devalued educational credentials, however, can be considered a complement to, not a substitute for, the trade, capital, and technology explanations. The question thus relies on its quantitative relevance, which might be relatively minor except for during some periods of employment contraction (for example, 1993–96).

Other studies have emphasized the link between adjustment, unemployment, and inequality by stressing macroeconomic factors rather than the credentialism hypothesis (González and Menéndez 2000; Altimir, Beccaria, and González 2002; Damill, Frenkel, and Maurizio 2003; and Beccaria 2006). However, as argued in the discussion that follows this, the *direct* distributional effect of the increased unemployment in the 1990s seems small, as it is accounted for mainly by the rise in labor market participation. The increase in unemployment may have depressed wages for the employed, especially among unskilled and semi-skilled workers, who bore the largest increase in joblessness, and that might have contributed to a higher wage premium and increased inequality. However, there is no systematic evidence on the strength of that phenomenon in Argentina.

In summary, some explanations stress the relevance of trade liberalization, technological change, and capital incorporation as the driving forces behind the increase in the relative demand for skilled labor. In contrast, the credentialism and macroeconomic adjustment arguments stress the negative impact of structural changes and the macroeconomic climate on the relative demand for unskilled labor. As argued above, the factors behind these explanations are not mutually exclusive and might even be complementary. That implies that the question of their relative merit needs to be answered empirically, although it is not easy to distinguish between the two sets of hypotheses with the data at hand.

Other Contributing Factors

Here we discuss factors that we did not cover previously, not because they are less noteworthy than the topics already presented but because data are lacking or identifying their effects is difficult. This short discussion will not do justice to these factors, but it provides some of the key references for further research.[13]

The first factor, demographic trends, had only a small impact on the income distribution. Marchionni and Gasparini (2007) develops a microdecomposition analysis that finds that a higher average number of children in middle- and low-income households in the period 1980–92 is associated with higher levels of poverty and inequality in equivalized household income. The same study shows

12. Gasparini and Cruces (2008) provides an alternative hypothesis for these findings.

13. A longer and more detailed discussion of these topics is presented in Gasparini and Cruces (2008).

that during 1992–98 household size decreased for most income groups, resulting in a small reduction in poverty and negligible effects on inequality.

Although less extensively reviewed than trade and capital account liberalization, further structural reforms occurred during the 1990s, such as privatization, decentralization, and labor deregulation. Several studies provide partial evidence of their distributive effects.

Until the beginning of the 1990s, the Argentine economy was characterized by the presence of large state-owned enterprises, especially in the utilities, oil, minerals, and financial sectors. In some sectors (especially utilities), privatization was successful in increasing efficiency and broadening service availability. However, its impact on prices means that the overall distributional effect of privatization has to be assessed on a case-by-case basis.

The main pro-poor argument in favor of privatization, the second factor, emphasizes the equalizing benefits from increased access to and supply of public services (Ennis and Pinto 2002). Galiani, Gertler, and Schargrodosky (2005) reports that during the 1990s municipalities that privatized water provision experienced a significant reduction in infant mortality rates (about 8 percent), which points to important equalizing effects in terms of welfare. On the other hand, the removal of implicit subsidies and investment finance resulting from privatization may generate higher prices for services, which could imply higher inequality. Ennis and Pinto (2002) reports that the share of a household's budget spent on telecommunications, natural gas, water, and electricity rose sharply after privatization and that the increases were significantly higher for households living below the median of the income distribution. Galiani and others (2004) argues that large efficiency gains from privatization did not translate into lower prices and that better regulation might have helped reduce prices in public utilities.

Given that most of the efficiency gains in privatized companies were achieved through layoffs, privatization also significantly affected Argentina's labor market. Privatization accounted for 13 percent of the increase in unemployment between 1987–90 and 1997. Layoffs affected as many as 67 percent of workers in the national railway company and 83 percent in Yacimientos Petrolíferos Fiscales (YPF), the state-owned oil company (Ennis and Pinto 2002). While the layoffs generated large losses for the affected workers, it is likely that they represented lower welfare losses for society as a whole. As noted in Galiani and Sturzenegger (2008), laid-off workers from YPF experienced a sharp fall in post-displacement earnings, which is compatible with the presence of rents related to employment in state-owned companies.

The decentralization of health services and secondary education (devolved from the central government to the provinces) was another important component of the 1990s reforms. Decentralization might affect human capital accumulation through health and educational outcomes; therefore, from a distributional perspective, it is related to the long-term distribution of well-being. Despite the arguments in favor

of decentralization, Cetrangolo and Gatto (2002) contends that in Argentina these reforms were driven by fiscal considerations. Likewise, they were usually rushed through without ensuring that provinces had sufficient funds and administrative capacity to provide quality health and education services. Galiani, Gertler, and Schargrodsky (2008) analyzes the 1992–94 transfer from the central government to provincial authorities of responsibility for all secondary schools. The authors find positive and significant effects of decentralization on standardized tests but that gains were concentrated in richer municipalities. That implies that this specific decentralization policy increased average quality but also increased the inequality in educational outcomes.

The third factor is labor unions, another factor in the labor market that has profound distributional implications. On one hand, unions tend to compress the distribution of wages within a sector; on the other hand, they also amplify the wage differentials between covered and uncovered workers (or insiders and outsiders). The Argentine labor market has been characterized by the presence of strong, industry-wide unions that played a significant role in shaping the country's social, economic, and political outlook, mainly through their relationship with the Peronist Party. Despite the importance of unions in the Argentine economy, there is little empirical evidence of their impact on wages and income, mostly because of lack of data.[14] While consistent time-series data on union membership are not available, the evolution of union coverage and strength from 1970 to 1983 can be deduced from qualitative sources (Marshall 2002). In broad terms, Argentina's unions were only somewhat weakened by the authoritarian governments at the beginning of the 1970s, but they were greatly weakened by the 1976–83 military regime. Unions regained substantial political and formal power with Argentina's return to democratic rule in 1983. According to data on strikes, union activity increased sharply from 1984 onward (Murillo 1997; Etchemendy and Collier 2007). The figures show a high degree of union activity and volatility during the 1980s; union activity then diminished greatly from 1991 to 1995 and grew again after 2001. Moreover, union membership also declined between 1990 and 2001 (Marshall 2005). The decline in union activity coincides with reforms such as privatization, trade liberalization, and price stabilization in the 1990s, which reduced the power of unions through the dissipation of rents from inefficient state-owned enterprises, protective tariffs, and inflation-induced rents. The decline in union activity during the 1990s thus coincided with a period of rising wage inequality and with factors that, according to the evidence reviewed, contributed to the rise in wage inequality. However, there is no estimate of the impact that a decline in union activity might have had on inequality.

14. The EPH, an otherwise fine labor force survey, has never routinely collected information on union membership.

Finally, it should be noted that from the mid-1990s a new form of organization emerged, representing the "outsiders"—or in more general terms, the disenfranchised—those who traditionally did not have union or political representation. Garay (2007) describes the surge of grassroots groups of unemployed and informal workers, which played an important role the late 1990s and in the establishment of the 2002 emergency cash transfer program, Programa Jefes y Jefas de Hogar Desocupados (PJJHD). Those organizations strengthened their power and gained political leverage, mostly through protest and through their ability to mobilize beneficiaries. By giving voice and representing the interests of those usually excluded, the groups tend to reduce their exclusion from the democratic process and thus to moderate political inequality.

The Distributional Impact of Government Social Policies

Although monetary income reported in household surveys includes some forms of monetary government transfers, the bulk of the government impact on individual well-being is ignored. In particular, the value of all in-kind subsidies, like education or health, is not included in the welfare proxy (household income) used for distributional analysis. The same is true for most of the tax system; monetary income includes only direct taxation, which has a minor role in Argentina, at least when compared with that in high-income OECD countries.

The results discussed here adjust inequality measures to account for the impact of social public policies, focusing on public benefits that are financed through general revenues and are not linked to contributions. The social policies analyzed include those concerning education, health, water, sanitation, poverty alleviation, housing, employment, and most municipal services (for example, street lighting and garbage collection).[15] Pension benefits are not included; because they are tightly linked to formal workers' contributions, the pension system is not a significant redistributive social public program. In theory, the exercise is simple: starting with the original income distribution reported in the survey, cash transfers are subtracted to get a "market" income distribution, to which the income value of in-kind social transfers is added and the burden of the tax system that finances social spending is subtracted.

The main results of the exercise are summarized in table 5-4. For spending, a higher concentration index means that more spending is directed toward higher-income people. Concentration indices for social public expenditures (SPE) are negative, indicating a mild pro-poor bias. Naturally, SPE is then progressive (column 4); the incidence of SPE is less concentrated on the rich than the income

15. See our companion paper (Gasparini and Cruces 2008) for details on the methodology and sources of information.

Table 5-4. *Distributional Impact of Social Spending and Taxes*

	Concentration index			*Kakwani progressivity index*				*Redistributive impact*	*Gini-post*
	SPE (1)	*Taxes* (2)	*Gini-pre* (3)	*SPE* (3) – (1) (4)	*Taxes* (2) – (3) (5)	*Total* (4) + (5) (6)	*SPE/DIa* (7)	(6) * (7) (8)	(3) – (8) (9)
1980	–0.144	0.361	0.391	0.535	–0.030	0.505	0.149	0.075	0.316
1981	–0.120	0.356	0.428	0.549	–0.072	0.476	0.143	0.068	0.360
1982	–0.116	0.362	0.421	0.537	–0.060	0.477	0.098	0.047	0.374
1983	–0.130	0.363	0.431	0.561	–0.068	0.493	0.118	0.058	0.373
1984	–0.131	0.351	0.440	0.572	–0.089	0.483	0.135	0.065	0.375
1985	–0.131	0.369	0.410	0.541	–0.040	0.501	0.141	0.070	0.339
1986	–0.139	0.366	0.420	0.559	–0.054	0.505	0.158	0.080	0.340
1987	–0.129	0.371	0.445	0.574	–0.074	0.500	0.168	0.084	0.361
1988	–0.115	0.374	0.454	0.569	–0.080	0.489	0.143	0.070	0.384
1989	–0.129	0.405	0.516	0.646	–0.111	0.534	0.126	0.068	0.449
1990	–0.115	0.376	0.462	0.577	–0.087	0.491	0.137	0.067	0.395
1991	–0.129	0.367	0.463	0.593	–0.096	0.496	0.147	0.073	0.390

1992	–0.122	0.340	0.452	0.574	–0.112	0.462	0.151	0.070	0.382
1993	–0.129	0.344	0.446	0.575	–0.102	0.473	0.169	0.080	0.366
1994	–0.133	0.350	0.455	0.588	–0.105	0.483	0.173	0.083	0.371
1995	–0.129	0.353	0.483	0.612	–0.130	0.482	0.168	0.081	0.402
1996	–0.135	0.358	0.487	0.622	–0.129	0.493	0.157	0.077	0.410
1997	–0.141	0.356	0.485	0.626	–0.129	0.497	0.160	0.080	0.405
1998	–0.138	0.362	0.504	0.642	–0.141	0.501	0.166	0.083	0.421
1999	–0.140	0.368	0.493	0.634	–0.125	0.509	0.184	0.094	0.400
2000	–0.142	0.379	0.507	0.648	–0.128	0.521	0.178	0.093	0.414
2001	–0.145	0.384	0.525	0.670	–0.141	0.529	0.188	0.099	0.425
2002	–0.172	0.391	0.554	0.725	–0.163	0.563	0.148	0.083	0.471
2003	–0.182	0.400	0.544	0.725	–0.144	0.582	0.141	0.082	0.461
2004	–0.175	0.390	0.518	0.693	–0.128	0.565	0.146	0.083	0.435
2005	–0.171	0.393	0.512	0.683	–0.119	0.564	0.158	0.089	0.423
2006	–0.165	0.392	0.493	0.659	–0.101	0.558	0.171	0.095	0.398

Source: Authors' calculations based on various sources (see text).

a. SPE/DI = Social public expenditure/Disposable income.

distribution. For taxes, a higher concentration index means higher-income people pay more taxes. Although taxes show a positive concentration index, according to column 5 they are not progressive. At least in this exercise, in which adjusted current (not permanent) income is the welfare proxy, the tax system, in contrast to SPE, is slightly regressive.

Two factors explain the increasing progressivity of SPE in Argentina. First, SPE's targeting of the poor has increased over time, especially since the implementation of large conditional cash transfer programs in 2002. However, Argentina's increasing income inequality is a second, unintended factor in the rising progressivity of SPE. If an income distribution becomes more unequal (as has been happening in Argentina), a fixed spending structure will become increasingly progressive.

Taxes are more concentrated in the upper-income strata, as reflected by their positive concentration index. Tax concentration rose in the 1990s but not at the same speed as income concentration, turning the tax system increasingly regressive, as shown by the increasingly negative progressivity index. That pattern changed in the 2000s as inequality started to decline after the 2001–02 crisis and significant changes in the tax structure favored the more progressive export and financial levies.

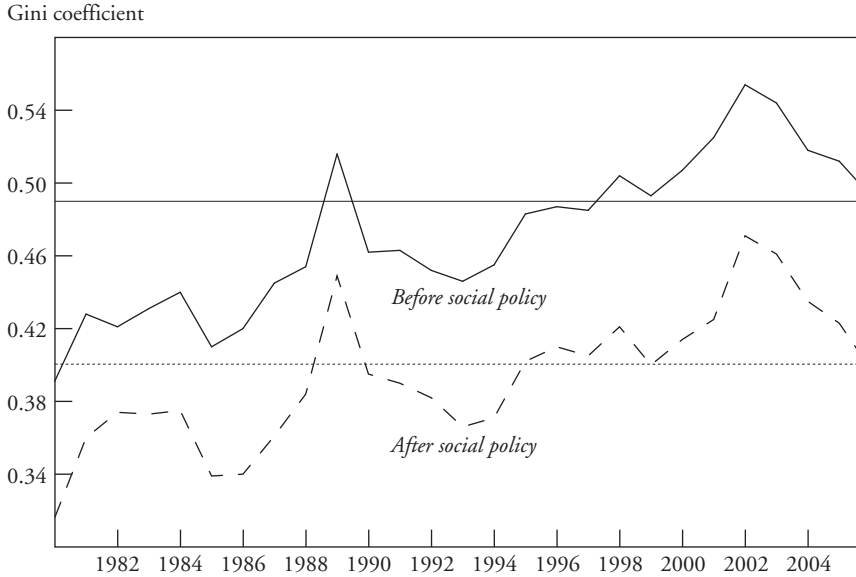
That point deserves further discussion. The government encouraged a large devaluation of the peso in 2002 and tried to sustain a higher exchange rate in the ensuing period. The devaluation implied a substantial fall in real wages and hence a potentially large impact in increasing inequality. On the other hand, the real devaluation of the peso, along with a surge in the price of agricultural commodities, created large rents to land. A fraction of those rents were extracted through progressive export taxes, which financed large poverty-alleviation programs. Therefore, the regressive devaluation was a key element that allowed Argentina to build and afford a substantially more equalizing fiscal system. It should also be noted that excise duties (or *retenciones*) had another indirect distributive impact. They kept local prices of traded goods below the international level, which was very important in a context of increasing international prices (the so-called commodity price boom). That effect is not included in these calculations, but it is discussed in Gasparini and Cruces (2008). Given the opposing effects in play, the net distributional impact of the peso devaluation is difficult to estimate.

Figure 5-4 shows that although Argentina's fiscal policy reduced inequality, it did not have a significant effect on the pattern of inequality over the last decades, primarily because the impact of fiscal policy on Argentina's income distribution was small compared with that of "market" forces on inequality.

The Impact of Macroeconomic Crises on Inequality

Unfortunately, macroeconomic crises have been a recurrent factor in the Argentine economy since at least the mid-1970s. The series in figure 5-2 depict sudden and large falls in GDP per capita associated with microeconomic crises. There

Figure 5-4. Gini Coefficient, Before and After Social Policy



Source: Authors' calculations.

seems to be a clear correlation between the evolution of GDP and distributional indicators during the episodes of crisis and recovery. There also appear to be sharp falls in inequality associated with macroeconomic stabilization. In what follows we analyze those trends and their potential determinants in Argentina.

The Crises

Several large crises occurred in Argentina in 1980–81, beginning with a banking crisis in 1980 followed by a large devaluation of the peso. In 1981, Argentina experienced a reversal in capital flows as a result of the collapse of a managed exchange rate system set up by the military government. The situation deteriorated further in 1982, when Mexico announced that it would default on its external debt. Mexico's default triggered an "external debt" crisis throughout the region. The devaluation of 1981 and the liquidity difficulties of 1982 (fueled also by the confidence crisis after the Malvinas conflict) resulted in a fall of around 5 percent in 1982 in Argentina's annual output. From 1980 to 1981, the Gini coefficient increased significantly (by about 3 points) and poverty rose by 2 percentage points. From 1981 to 1982, the Gini fell almost 1 point. The poverty headcount ratio, however, continued to increase by 3 percentage points during the year; by 1982 it rose above 10 percent.

Argentina's second big crisis took place in the mid-1980s. It began with the 1985 recession, which preceded the Austral stabilization plan. Although the

Austral stabilization plan initially managed to reduce inflation, the effects were not permanent. Inflation accelerated again during 1985–88, culminating in two hyperinflation attacks: one in mid-1989 and one in the first quarter of 1990. Output fell 11 percent between 1988 and 1990, and annual inflation rates were 343 percent in 1988, 3080 percent in 1989, and 2314 percent in 1990. Hyperinflation had a very large distributional impact: poverty increased by 25 percentage points and the Gini coefficient increased by 6.3 points between 1988 and 1989 (although it should be noted that inequality had been increasing steadily between the 1985 stabilization and the 1989 hyperinflation-induced jump).

In April 1991, the country adopted a currency board with a fixed exchange rate regime, the convertibility plan, which managed to curb inflation successfully. The convertibility plan was accompanied by a series of far-reaching structural reforms. The Argentina economy grew fast, fueled by growing public and private indebtedness from the liquidity in international capital markets, until 1994. A currency crisis in Mexico at the end of 1994 triggered capital flight from the region and a financial crisis that severely affected the Argentine economy. The currency board sustained the drain of reserves, but the so-called "Tequila crisis" implied a fall in Argentina's GDP of around 4 percent in 1995. Argentina's growth bounced back quickly from the contagion of the Mexican financial crisis. Nonetheless, the Tequila crisis had a large distributional impact: poverty increased by 5 percentage points and the Gini went up by 2.7 points in a year.

The effects of the crisis are also visible in the national urban unemployment rate, which soared from 10.7 in May 1994 to 18.4 percent in May 1995. The hike was probably partially a consequence of the reforms; it already had been increasing steadily, from 6 percent in October 1991 (the first measure before the implementation of the convertibility plan). But unemployment jumped with the Tequila crisis and remained in double digits until the second half of 2006. The crisis also generated a set of changes in the structure of employment, the most notable being an increase in the labor force participation of women and secondary workers. A second change was the implementation of the Plan Trabajar, a workfare program, which would form the basis for future cash transfer initiatives. However, neither inequality nor the unemployment rate fell during the recovery, as they did in the period after the hyperinflation crisis.

It is possible that the financial crisis acted as a catalyst that accelerated and amplified the adverse effects of the reforms, which were bound to produce large sectoral reallocations of employment. Households and small firms were especially vulnerable in the context of an overvalued exchange rate, increased competition, and almost no compensatory policies to ease the transition (Plan Trabajar was implemented after the crisis, not during the initial reforms). The Tequila crisis probably exacerbated what was going to be, in any case, a difficult transition. While the crisis had long-lasting effects on the income distribution (and on

unemployment levels), it is likely that they were partially due to the acceleration and worsening of a process that was already under way.

While growth resumed fairly strongly from 1996 to 1998, inequality fell only slightly. The continuing exposure to international capital flows brought about by the currency board regime and capital account liberalization hit the Argentine economy at the end of the millennium. As a result of the 1997 financial crisis in Southeast Asia and the 1998 crisis in Russia, in January 1999 Brazil, Argentina's largest trading partner, devalued its currency. The Argentine economy entered a period of recession, which culminated in a major economic, banking, and financial crisis. In December 2001, the currency board system collapsed, after restrictions were imposed on withdrawal of funds from the banking system. From 2000 to 2002, Argentina's GDP per capita fell 17 percent. The difference is even greater when comparing 2002 with 1998, when the recession started; over that five-year period Argentina's GDP per capita fell 22 percent.

The recession and the subsequent economic crisis in 2001 had a large impact on poverty, which rose 26 percentage points between 1999 and 2002. It also had a significant negative effect on inequality: the Gini coefficient increased 4 points from 1999 to 2002. The most dramatic effect was the combination of increasing prices (due to the devaluation) and falling nominal incomes (due to the sharp fall in economic activity), which instigated a jump in Argentina's official poverty rate from 38.3 percent in October 2001 to 53 percent in May 2002. Nonetheless, by 2003 the economy began to recover, and from 2003 to 2007 it grew at very high rates (above 8 percent per year).

Large macroeconomic crises: the impacts of hyperinflation on income distribution. From the macroeconomic-distribution perspective, the two largest crises (1989 and 2001–02) represent the most interesting episodes in the years under study. Both crises precipitated unusually large falls in GDP and simultaneous large negative impacts on Argentina's income distribution.

Given the fiscal origin of most bouts of high inflation and hyperinflation (Heymann and Leijonhufvud 1995), it is not surprising that the literature concentrates on the differential incidence of the inflation tax.[16] Ahumada and others (1993) and Canavese, Sosa Escudero, and González Alvaredo (1999) quantify the distributional effects of inflation in Argentina from a tax-incidence perspective. Both studies estimate Argentina's aggregate monetary demand functions and aggregate inflation tax and seigniorage collection and include a re-weighted consumer price index by quintile. Canavese, Sosa Escudero, and González Alvaredo (1999) finds that, except for the hyperinflation during 1989 and 2001–02, quintile-specific inflation rates did not differ significantly over time. However, from 1980 to 1990, the infla-

16. Inflation constitutes by definition a proportional tax on nominal balances, but its effect on the income distribution is neutral only if all households face the same inflation rate, if all households have the same income elasticity in their demand for money, or if they all have access to the same "protection" technologies, such as currency substitution or indexed contracts.

tion tax as a proportion of income was about twice as large for households in the poorer quintile as for those in the richer quintile. The impact of the inflation tax on aggregate inequality indicators was comparatively small for high-inflation periods, with increases of about 1 to 1.5 points in the Gini for 1980–88 and 1990. In contrast, in 1989 the inflation tax produced an increase of 3.4 points in the Gini coefficient. The inequality-increasing impact of hyperinflation, however, was short lived. Stabilization programs reduced the inflation tax, as witnessed by the 1991–93 downward trend in the Gini coefficient (figure 5-2).

Large macroeconomic crises: the distributive effects of the 2001–02 meltdown. The crisis induced by the implosion of the currency board regime in December 2001 was unusually virulent, even by Argentine standards. The origins and consequences of the crisis have been analyzed extensively, and a selection of the existing studies is covered in the following discussion.

Using a survey implemented by the World Bank in the midst of the crisis (June and July 2002), Fiszbein, Giovagnoli, and Adúriz (2003) found that almost half of households suffered a fall in nominal income. They also documented a change in household roles, with respect to the labor market, with higher employment among secondary workers as a strategy to complement the fall in income from primary workers who were unemployed or working reduced hours. As in other crises in Latin America, the extremely high level of unemployment meant that school enrollment did not fall significantly among younger children and only slightly among those aged 16 to 18 years.

One key component of the crisis was a large bank deposit freeze, which in principle has an ambiguous direct distributional effect.[17] Halac and Schmukler (2004) finds that the probability of having savings was positively and significantly associated with measures of income. Interestingly, however, the authors found that among those with savings, the less educated and those with lower incomes had a larger probability of being affected by the bank deposit freeze. That finding suggests that the measure probably did have an inequality-increasing effect.

McKenzie (2004) examines the labor market responses to the 2001-02 crisis based on the panel structure of the EPH. They found that most of the fall in households' real income was due to the fall in the real wages of individuals remaining in the same jobs. Because unemployment was already at very high levels from the 1999–01 recession, it only rose 3 percentage points during the crisis. Cruces (2005), using the same panel feature of the EPH data, found that households exposed to the 2001–02 crisis experienced significant increases in income variability (70 percent).

Permanent effects of the crises. The available evidence clearly shows that in Argentina the poor were more affected by crises than the non-poor. How-

17. While no empirical analysis has attempted to link the two phenomena, it is widely believed that the restrictions on withdrawing cash from banks had a poverty- and inequality-increasing effect by starving the cash (or informal) economy.

ever, most of the inequality-increasing factors mentioned here tend to dissipate relatively quickly through increases in employment and income levels as the economy recovers. In the two large crises, inequality first jumped but then fell drastically right after economic stabilization. There is a debate on the existence of hysteresis effects on inequality from the crises (Lustig 2000), but there does not seem to be definitive empirical evidence for Argentina. While the Tequila crisis had permanent effects on the income distribution, it is likely that they were due to the acceleration of the negative aspects of the underlying reform process. Moreover, any permanent effects of the 2001–02 crisis are difficult to evaluate, since they are confounded with the strong recovery from 2003 onward. Given the aforementioned evidence that school enrollment did not decline during the crisis, the typical argument of a permanent reduction in the stock of general human capital does not seem to apply for the 2001–02 crisis. However, other more subtle mechanisms might have influenced inequality, like losses of firm-specific human capital for the newly unemployed, who were for the most part unskilled workers. Those impacts, and the psychological costs of long-term unemployment, are some of the questions on the impact of the 2001–02 crisis that merit further research.

The Fall in Inequality during the Post-Crisis Recovery: 2003–07

A new episode unfolded in Argentina after the peak of the crisis in 2002. Between 2003 and 2007, the average annual growth rate was an unprecedentedly high 8 percent and the unemployment rate plummeted from more than 20 percent to 8 percent. Inequality fell sharply from 2003, and by 2006 it had descended to approximately the same level as in the second half of the 1990s, before the start of the 1999–2001 recession and the subsequent crisis.

The strong devaluation of the peso in 2002 generated a new structure of relative prices that propitiated the economic recovery. The fall in real wages increased the competitiveness of Argentina's products and deterred imports. New taxes and a default on the government's debt obligations allowed for a fiscal surplus that helped stabilize the economy. The new and stronger government from the traditional Peronist Party (2002–03) curbed the social unrest and the political instability of 2001–02 with the help of a large cash transfer program, which achieved unprecedented coverage. Moreover, Argentina saw a large increase in the prices of commodities that it exported during those years, and the exceptional conditions in the international markets also were a key factor in the recovery.

Reasons for the fall in inequality

Several factors combined to create a scenario in which inequality fell: the recovery from the crisis of 2001–02; realignments after the devaluation of the peso; strong expansion of employment; productive changes after new relative prices;

slower technical upgrading; stronger labor institutions and labor policies; and a more extensive safety net.

As discussed previously, major macroeconomic crises, which seriously disrupt the economy, are associated with large jumps in inequality. When economic life returns to some level of normality, inequality always falls. Although the 2001–02 crisis was considered one of the worst in Argentina's history, by 2004 the economy had fully recovered, with GDP levels similar to those in pre-crisis years. Some of the drop in inequality after 2002 was driven by the recovery and rebuilding of business and economic relationships after such a large shock.

After years of pegging the Argentine peso to the dollar at a clearly overvalued rate, in January 2002 the currency board regime was abandoned, and as a result the nominal exchange rate increased threefold in only a few months. That dramatic devaluation implied a sudden fall in real wages, and hence inequality rose. As in many devaluation episodes, the full impact of the shock in 2002 started to dissipate as nominal wages increased, and the nominal exchange rate recovered from its initial "overshooting." The increase in nominal wages originated from workers' demands to recover part of the large real losses in purchasing power due to the devaluation. A substantial part of the fall in inequality since 2002 can be attributed to those realignments.

The strong recovery of the economy meant higher demand in a labor market with depressed real wages. The share of adults employed jumped from 35 percent in 2001 to 43 percent in 2006. Accordingly, the unemployment rate fell from around 20 percent in 2001–02 to 10 percent in 2006. By 2006 unemployment had finally declined to a level lower than in most of the 1990s. The expansion in employment was surely another contributing factor to the fall in inequality. Employment affected inequality through both the direct channel of rising incomes for the formerly non-employed and the wage-increasing pressures on the labor market.

The change in relative prices due to the devaluation benefited industries that relied on unskilled labor (such as the textile industry), which had faced strong import competition under the previously overvalued exchange rate. For the reasons discussed previously, it is likely that the trade channel (through import substitution) also contributed to lower inequality. Unfortunately, the evidence on the distributive effects of trade-induced changes is fragmentary.

As argued above, the Argentine economy in the 1990s experienced substantial changes in the organization of production. In the 2000s, the rate of technology adoption and capital investment is believed to have been slower than during the 1990s for several reasons. A large upgrading increase already had taken place in the previous decade. Firms also resorted to idle capacity after the crisis. Finally, investment might have slowed because of the uncertainty of the social unrest brought about by the crisis, because of an atmosphere of distrust following the default on the country's debt, and because of some perceptions that the government was send-

ing signals that were unfriendly to markets. The literature typically finds inequality overshooting after skill-biased technical shocks because it takes time for displaced unskilled workers to be reallocated in the economy,[18] and the initially strong unequalizing impact of the technological shock of the 1990s should have lost strength over time. That adjustment after the production shock was interrupted by the recession of the late 1990s and the crisis of the early 2000s, but it likely resumed during the recovery, with the ensuing equalizing consequences.

Beginning in 2002, Argentina's governments took a more active role in the labor market by increasing the minimum wage, mandating lump-sum increases in wages (which compresses their distribution), taking a pro-union stance, and promoting collective bargaining. The government also implemented active policies, ranging from subsidies to firms to export taxes on agricultural products, to keep the prices of utilities and foodstuffs low. While export taxes that introduce a gap between domestic and international prices have all sorts of leakages to the non-poor and inefficiency costs, in the short run they have an equalizing effect given the higher share of foodstuff in the consumption bundle of the poor.

As discussed above, the economic expansion and new price structure improved the redistributive impact of fiscal policy. Social spending rose, taxes became somewhat more progressive (mostly because of export levies), and the increased revenues went in part to sustain a large system of pro-poor cash transfers. The Programa Jefes y Jefas de Hogar Desocupados, which covered 2 million households in 2003, was still running in 2007, and new cash transfer programs were being implemented. The impact of the PJJHD lost strength as the economy recovered because beneficiaries left the program to work but also because the cash subsidy remained fixed in nominal terms in a context of increasing inflation.

Finally, it is interesting to notice that although inequality fell substantially with respect to the crisis levels, inequality in 2006 was not significantly different from that of the mid- to late 1990s, despite the fact that per capita GDP and employment were higher, labor institutions were stronger, and a massive cash transfer program was implemented. That rigidity is probably due, at least in part, to the fall in the relative productivity of unskilled workers that followed the modernization of the economy in the 1990s. The main challenge for distributive policies in the long run in Argentina is to raise the productivity of this group of workers and of their offspring, allowing them to share the benefits of economic growth.

Concluding Remarks

This chapter describes the level and evolution of inequality in Argentina from the mid-1970s to the mid-2000s by reviewing the empirical evidence behind a series of determinants of that evolution. Along the way, it also attempts to pinpoint

18. See, for example, the discussion by Jaime Kahhat in chapter 2 of this volume.

which of the factors contributing to the dynamics of inequality can be attributed to market forces, to state action, to sociopolitical influences, or to different combinations of those elements.

Only time will tell whether the reforms implemented will have a positive impact on welfare through the growth channel and whether a trade-off exists between short-term inequality and long-term growth. Too many factors are in play to construct clean counterfactuals, but the evidence on the poverty- and inequality-reducing effects of government action seems to suggest an excessive increase in inequality after macroeconomic shocks and reforms, with real welfare costs. Those costs might have been mitigated by the presence of ex-ante policies for coping with shocks and by compensation and income-smoothing mechanisms to ease the transitions after reforms.

Some of the factors that account for the long-term trends in inequality can be attributed to government acts and others to market forces, but most of the distributional factors covered in this chapter correspond to a mix of both that might be ultimately impossible to disentangle—for instance, trade liberalization corresponds to the operation of market forces following a change in government policy, while most other changes occur within the government's regulation of an economy. The evidence, however, has shown that government intervention can have powerful effects on the distribution of welfare, and the government role in managing and compensating losers during transitions must not be underestimated.

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DECLINING INEQUALITY IN LATIN AMERICA

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editors

UNITED NATIONS DEVELOPMENT PROGRAMME
New York

BROOKINGS INSTITUTION PRESS
Washington, D.C.

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1775 Massachusetts Avenue, N.W., Washington, DC 20036
www.brookings.edu

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Library of Congress Cataloging-in-Publication data

TK

9 8 7 6 5 4 3 2 1

Printed on acid-free paper

Typeset in Adobe Garamond

Composition by R. Lynn Rivenbark
Macon, Georgia

Printed by R. R. Donnelley
Harrisonburg, Virginia.